HDOC DAY – 67

Algorithm

1. Start.
2. Read the decimal number n.
3. If n is 0, display 0.
4. Otherwise, call decimal_to_binary(n).
5. If n = 0, return from the recursive function.
6. Recursively call the function with n / 2.
7. Display the remainder n % 2.
8. Continue until all binary digits are displayed.
9. Stop.

C PROGRAM:

```
#include <stdio.h>


void decimal_to_binary(int n)
{
   if (n == 0)
      return;


   decimal_to_binary(n / 2);


   printf("%d", n % 2);
}


int main()
{
   int n;
```

```
    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary representation: ");

    if (n == 0)
        printf("0");
    else
        decimal_to_binary(n);

    printf("\n");

    return 0;
}
```

TEST CASE TABLE:

Test Case	Decimal Number	Expected Output
1	10	Binary representation: 1010
2	5	Binary representation: 101
3	8	Binary representation: 1000
4	15	Binary representation: 1111
5	0	Binary representation: 0